

Support-aware training of hard Top- K under the code residual without a surrogate backward

8-layer TinyStories models of the K+J notes, `code_residual`, abs-TopK over $N = 1536$; runs `ma_alt_*` (madrid), configs `configs/cr_kj/alt/`.

1 Question

The K+J notes ask for a hard Top- K model whose training is aware of other supports: candidates just outside the support should receive a signal about whether they belong in it. RBLapSum supplies that signal as a surrogate backward – a relaxed Jacobian $M + S$ of the gate, with S the kernel-weighted exchange term over the Top- $(K+J)$ pool – and under the code residual, where the gate’s output is also a skip connection, the relaxed Jacobians of successive gates multiply along the carry (`docs/carry-surrogate.tex`). Fixing that by re-routing gradients is a repair of a surrogate that does not fit the architecture. This note asks the question the other way round: which training rules give the candidates a signal *without* a surrogate backward, so that nothing can compound because every backward is the exact gradient of some forward?

2 Three candidates

A. Stochastic support width (`stochastic_width`). Every forward is a hard Top- K' forward; per token, K' is drawn from a distribution on $\{K, \dots, K+J\}$ in training and is K at evaluation. The objective is $\mathbb{E}_{K'} \mathcal{L}(K')$ and the gradient is exact; a candidate at rank $K+m$ is active, and trained, with probability $P(K' \geq K+m)$, so the distribution on K' is the probability distribution on the pool, realized in the forward instead of in the backward. Three distributions: uniform on $[K, K+J]$; two-point ($K+J$ with probability $\frac{1}{2}$, else K); geometric excess with mean $J/4$. This is the width sampling of universally slimmable networks (Yu & Huang 2019) and the nested training of Matryoshka SAEs (Bussmann et al. 2025) applied to the support, with the smallest width as the deployed one. Under the code residual the carried code is hard at every width, so the carry is exact by construction.

B. Soft straight-through (`surrogate_mode=soft_ste`). Hard forward; backward Jacobian $M + (1 - M)p$ over the pool, $p_i = F(|u_i| - b)/T$ the kernel’s CDF: a candidate receives the gradient it would receive if present, weighted by its probability of being present. No exchange term. Every per-coordinate gain is at most one, so the products along the carry are bounded by one; the price is that the candidates’ values are trained on a counterfactual, as in any straight-through estimator (Bengio et al. 2013; JumpReLU SAEs, Rajamanoharan et al. 2024).

C. RBLapSum on the candidates only (`rblapsum_surrogate_scope=inactive`). The support term is applied to the J inactive members and the K active ones keep the exact gradient. The argument: the active values already have a gradient, and if the loss wants one smaller it shrinks and leaves; only the entry decision is blind. A candidate is not carried (its code value is zero), so the term is applied once per entry event and cannot compound along the identity.

3 Results

Setup. The code-carried cells of the K+J grid, 3000-step screens with the grid’s 20k schedule untouched, step-matched against the grid’s hard Top- K baselines (post-norm) and its RBLapSum `pool` cells, with the carry scopes of `docs/carry-surrogate.tex` for reference. A trains the hard baseline’s config (post-norm) with a random width; B and C train the `pool` cell’s config (no post-norm, $T = 2$) with one field changed.

At step 3000, minus hard Top- K (Fig. 1, Table 1). At (256, 256), the cell where the unmodified surrogate loses 0.047, every width distribution except the narrowest beats the hard model – two-point -0.008 and -0.009 ($p = \frac{1}{2}$, $\frac{1}{4}$), uniform -0.006 , geometric $J/4 -0.005$, geometric $J/16$ 0.000 – with no surrogate and no gradient modification, as much as the best carry scope (-0.005); candidates-only RBLapSum ends at $+0.004$, the soft straight-through at $+0.030$. At (128, 128) the widths give -0.012 (geometric $J/4$), -0.010 (two-point) and -0.008 (two-point $p = \frac{1}{4}$), against -0.006 for the mixed carry scope and -0.007 for candidates-only RBLapSum (`pool` $+0.025$). At (32, 224) the order of the width distributions follows how much mass they put near K : uniform $+0.014$, two-point -0.019 at both $p = \frac{1}{2}$ and $\frac{1}{4}$, geometric $J/16 -0.025$, geometric $J/4 -0.034$; at (64, 64) the two-point and geometric widths

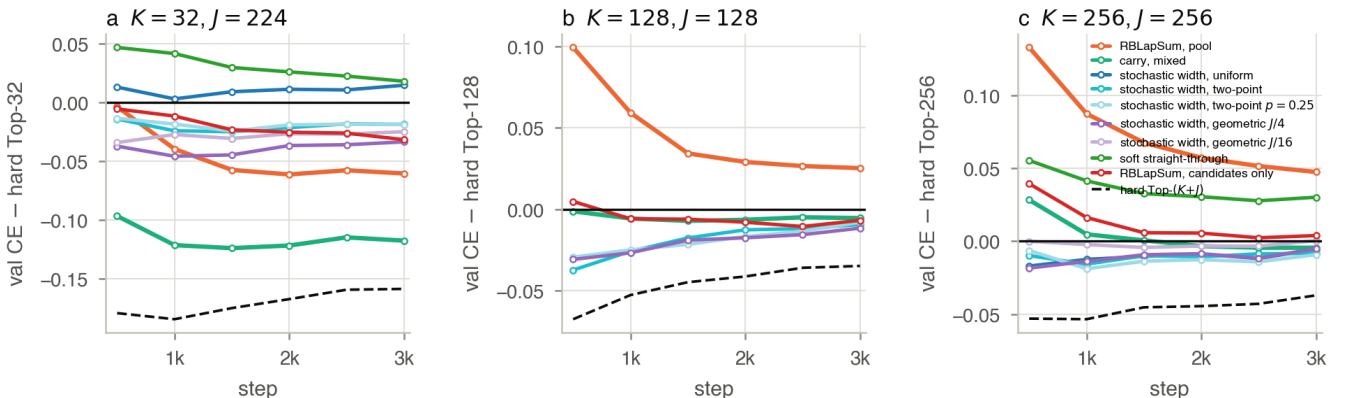


Figure 1: Validation CE minus the code-carried hard Top- K baseline over the first 3000 steps; dashed, hard Top- $(K+J)$. **a** (32, 224); **b** (128, 128); **c** (256, 256).

K	J	pool	carry mixed	stochastic width				soft STE	cand. only
				uniform	two-point	geom. $J/4$	geom. $J/16$		
32	224	-0.061	- 0.118	+0.014	-0.019	-0.034	-0.025	+0.018	-0.032
64	64	-0.005	- 0.047	-	-0.029	-0.023	-	-	-
128	128	+0.025	-0.006	-	-0.010	- 0.012	-	-	-0.007
256	256	+0.047	-0.005	-0.006	- 0.008	-0.005	0.000	+0.030	+0.004

Table 1: Validation CE at step 3000 minus the code-carried hard Top- K of the row (1.899, 1.835, 1.776, 1.741); two-point with $p = \frac{1}{2}$. Bold: best of the row.

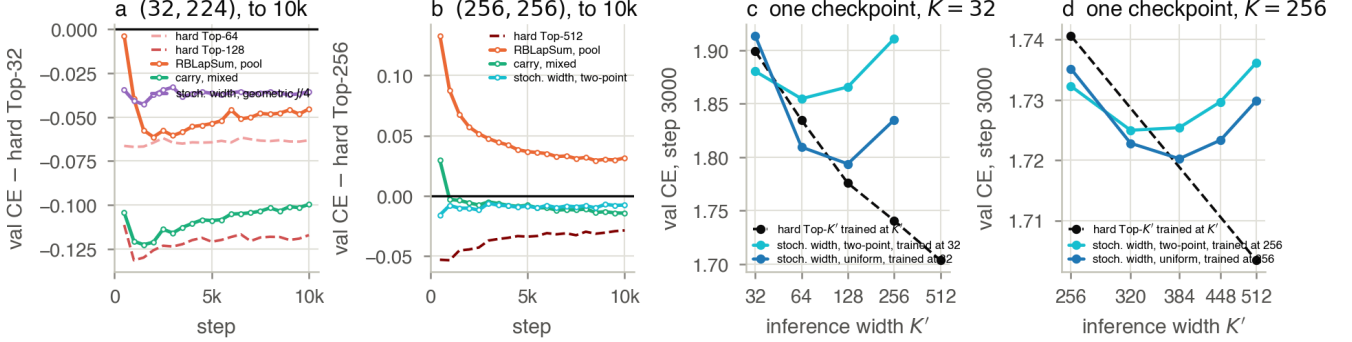


Figure 2: a, b The 10k continuations: the best stochastic width of each cell (geometric $J/4$ at (32, 224), two-point at (256, 256)) against the mixed carry scope’s continuation, the unmodified surrogate and the hard ladder. c, d One checkpoint, several inference widths: validation CE at step 3000 of the stochastic-width models evaluated with hard Top- K' across their training range, against the grid’s hard models trained at each K' (analysis/eval_widths.py).

give -0.029 and -0.023 (pool -0.005, mixed carry scope -0.047, hard Top-128 -0.059); candidates-only RBLapSum -0.032, soft straight-through +0.018, the unmodified surrogate -0.061, the mixed carry scope -0.118. Continued to 10k at (256, 256) (Fig. 2b), the two-point width holds its margin, -0.008 at every point from 4k on, while the mixed carry scope’s grows from -0.005 to -0.014 and overtakes it at 6k; hard Top-512 is at -0.029. At (32, 224) the geometric width holds -0.033 to -0.041 at every point to 10k (-0.036 at 10k), while the unmodified surrogate decays from -0.061 to -0.045 and the mixed carry scope from -0.118 to -0.100; hard Top-64 is at -0.063 (Fig. 2a).

What a stochastic-width model is (Fig. 2c,d). Evaluated across its training range, the model is not a slimmable ladder: its loss is lowest at an intermediate width and rises again toward $K+J$, where it is far above the hard model trained there (1.736 against 1.703 at $K' = 512$). At the narrow end it beats the hard models (the uniform model trained at $K = 32$ gives 1.809 at $K' = 64$ against 1.835 for hard Top-64). The random width acts as a regularizer on the support – the network cannot rely on any particular candidate being present – not as a way of training several models at once.

4 Reading

The probability distribution on the pool can live in the forward. RBLapSum puts a distribution on the Top- $(K+J)$ pool and differentiates through it; under the carry that derivative compounds because the same carried value is relaxed at every gate. Sampling the width puts the distribution on the pool into the forward instead – a candidate at rank $K+m$ is present with probability $P(K' \geq K+m)$ and, when present, is trained by the exact gradient of a hard forward – so there is no surrogate Jacobian, nothing to route, and the carried code is a hard code at every step. At (256, 256) this is enough: every width distribution tried beats the hard model (-0.005 to -0.008), the carry scopes’ result without the carry scopes. The price is a train/test mismatch on the support width: the deployed model has seen its own width only part of the time, and the gap to the hard baseline at $K = 32$ depends on how much of the distribution sits at K (uniform +0.014, two-point -0.019, geometric $J/16$ -0.025, geometric $J/4$ -0.034). Where the unmodified surrogate is worth 0.06 and the mixed carry scope 0.12, the best width distribution tried is worth 0.034.

What the hard forward cannot give the candidates. The two surrogate-free rules differ in what a candidate learns from. Under the stochastic width it learns from being *present*, in a forward where the whole stack adapts to its presence; under the soft straight-through it learns from a counterfactual, the gradient it would receive if present, with the rest of the stack unchanged. The second is worse than the hard backward at both K (+0.018, +0.030), while RBLapSum restricted to the same candidates with its boundary exchange term (-0.032, +0.004) is not: the value of RBLapSum’s signal is the *relative* ranking of candidates near the boundary, not the counterfactual gradient as such.

Against the carry scopes. The carry scopes keep RBLapSum’s signal, including the eviction term that the measurements of docs/carry-surrogate.tex found decisive at $K = 32$, and make it compatible with the carry by a rule about where the kernel acts. The stochastic width is the natural surrogate-free training of a hard Top- K under the code residual and matches them at large K , where the hard model’s own selection is already good and the surrogate’s job is only to keep candidates alive; at small K , where the selection itself is worth 0.1 nats, a distribution over widths does not reach what a gradient on the ranking does, although its margin does not decay with training where the surrogates’ do (-0.036 flat to 10k against -0.118 $\rightarrow$ -0.100 and -0.061 $\rightarrow$ -0.045).

Scope. One seed, 3000-step screens and two 10k continuations in the $T = 2$ code-carried family; the stochastic-width models carry the hard baselines’ post-norm, the surrogate cells do not (about 0.007).